Report on *"Matrix Theory Reloaded: A BPS Road to Holography"*

(Blair, Lahnsteiner, Obers, Yan)

Summary. The paper takes the BPS decoupling limit that produces matrix theory on Dp-branes ("matrix p-brane theory", MpT) and applies it to all of type II string theory, so that the target space becomes a codimension-(p+1) foliated, non-Lorentzian geometry. It then makes four claims. (i) Applying this limit at asymptotic infinity of a Dp-brane solution produces exactly the Maldacena/IMSY near-horizon limit in the bulk. (ii) Repeating the limit for misaligned branes generates non-Lorentzian holographic pairs, including those of Lambert et al. (iii) All of these can be organised as a "DLCQ^n/DLCQ^m" correspondence. (iv) Running the limit backwards is a p-brane generalisation of $T\bar{T}$, which yields explicit flow equations taking free fields (or Maxwell) to Dirac–Nambu–Goto (or DBI).

1. Principal achievement

The main achievement is a single *generating rule* for decoupling limits on both sides of holography. The bulk near-horizon limit and the field-theory decoupling limit turn out to be the *same* BPS limit, seen in different regions of one supergravity solution (Sec. 3.1). Running the logic backwards, the emergent dilatation symmetry of MpT lets the limit parameter ω be promoted to $\Omega(r) = (r/\ell)^{(7-p)/2}$. The near-horizon geometry then reads as a "position-dependent" MpT limit, which explains why its asymptotic region is the flat non-Lorentzian MpT geometry seen by SYM.

For $p = q$ this is largely a geometric repackaging of known physics, which the authors acknowledge. Its value is that it is *algorithmic*:

- The condition $m + n = 4$ (relating finite-mass branes, finite RR potentials and 1/4-BPS intersections) gives the catalogue of Dq/MpT backgrounds in Table 1. This reproduces and extends the non-Lorentzian proposals in the literature.
- The D1–D5 system cleanly shows how intersecting branes fit in.

The most concrete, independently checkable output is the general flow equation of Sec. 6:

$$\partial_t \mathcal{L} = \tfrac{1}{2t^2}\Big\{\mathrm{tr}(1 - t\mathcal{T}) - (p-1)\big[\det(1 - t\mathcal{T})\big]^{\frac{1}{p-1}} - 2\Big\}.$$

It reproduces $T\bar{T}$ at $p = 1$ and the Gross et al. quantum-mechanics deformation at $p = 0$. For $p = 2$ it gives a strikingly simple membrane flow. For $p = 1$, its F-dependent version explains why ordinary $T\bar{T}$ does not deform 2d Maxwell into DBI.

I checked this equation numerically: for $p = 0, \ldots, 5$, with several scalars and a constant curved $\tau_{\alpha\beta}$, it holds to $\sim 10^{-8}$. I also checked its $p = 2$ reduction analytically.

2. Principal drawback

The paper is long on organising principles and short on *dynamical tests*. Its central conjecture, "holography is a DLCQ^n/DLCQ^m correspondence", is so far a way of *counting limits*. It makes no new prediction that could fail. No observable (spectrum, correlator, index, entropy, or even the isometry superalgebra) is computed on either side of any new non-Lorentzian pair.

The supporting bulk geometries also rest on shaky ground:

- The authors concede (Sec. 7) that they do not check the MpT supergravity equations or the torsion constraints.
- Most entries of Table 1 need either smearing or an *ad hoc* rescaling $N \to \omega^{3-q+m}N$, whose physical meaning is left open.
- The DLCQ^3 assignment for Spin Matrix Theory is speculative.

The "intrinsic generation of the bulk via p-brane $T\bar{T}$" (Sec. 5.4) is also weaker than advertised. The profile $\Omega(r)$ is put in by hand, using the dilatation symmetry; it is not derived from the flow, and fixing it still needs the bulk equations of motion. So, as stated, the construction is kinematic.

Finally, the new flow equations are classical, bosonic and abelian, and for $p \geq 3$ with a gauge field they are known only perturbatively. For $p > 1$ the deforming operator is non-polynomial in $T_{\alpha\beta}$ and depends explicitly on t. Nothing is said about whether it can be defined quantum mechanically (point splitting or factorisation), and that is exactly what makes $T\bar{T}$ special in 2d.

3. Most promising follow-up (feasibility vs. impact)

Make the simplest DLCQ^1/DLCQ^2 example quantitative by uplifting it to null-compactified AdS$_7 \times S^4$. The cleanest new pair in the paper is the M0T limit of the near-horizon D4 geometry (Sec. 3.2, with $p = 0$, $q = 4$), proposed to be dual to 5d SYM localised on instanton moduli space.

A short computation shows that this geometry lifts *exactly* to known 11d physics. Use the DLCQ uplift of Sec. 4.1, $\tau^{(11)} = e^{\varphi/3}\tau$ and $h^{(11)} = e^{-2\varphi/3}E$, with $\tau = (r/\ell)^{3/4}dt$, $e^{\varphi} = (r/\ell)^{3/4}$ and $E = (r/\ell)^{3/2}dx^I dx^I + (\ell/r)^{3/2}(dr^2 + r^2 d\Omega_4^2)$. This gives

$$ds_{11}^2 = \tfrac{r}{\ell}\big(2\,dx^+dx^- + dx^I dx^I\big) + \big(\tfrac{\ell}{r}\big)^2\big(dr^2 + r^2 d\Omega_4^2\big), \qquad x^- \sim x^- + 2\pi R,$$

i.e. the M5-brane near-horizon geometry AdS$_7 \times S^4$ with one null boundary direction compactified. The boundary dual is then the DLCQ of the $(2,0)$ theory, which is the known instanton-moduli-space quantum mechanics (Aharony–Berkooz–Kachru–Seiberg–Silverstein), at null momentum K.

This suggests a concrete programme:

1. *Solve the open consistency question for this example.* The M0T background is automatically a solution, since it is a null quotient of an exact 11d solution. Use it to derive the M0T (non-Lorentzian IIA) field equations and torsion constraints by expanding in ω, and check them against this solution.
2. *Symmetries.* The isometry superalgebra is the part of $\mathfrak{osp}(8^*|4)$ that commutes with P_-, a Schrödinger-type superalgebra. Match it to the symmetries of the Lambert et al. field theory. The paper itself flags the isometry group as "natural and important".

3. *A real test.* Compare the large-N spectrum of protected states at fixed K: KK supergravitons on $\mathrm{AdS}_7 \times S^4$ with $p_- = K/R$, against the Hilbert series/index of the K-instanton moduli space. Then compute two-point functions from the bulk-to-boundary propagator on the non-Lorentzian geometry, written directly in M0T variables.

4. *Extend and stress-test.* The smeared entries of Table 1 are T-dual to this case, so they inherit the result. Comparing with the large-N-rescaled versions would show which of the paper's two inequivalent families is the physically relevant one. Steps 3–4 would also expose whether the known subtleties of null quotients (the $p_- = 0$ sector, strong coupling near the horizon) are controlled better in M0T variables, as the paper implicitly claims.

Why this one. Every ingredient (11d supergravity on $\mathrm{AdS}_7 \times S^4$, instanton moduli-space indices, Schrödinger holography) is existing technology, so step 1 is short and steps 2–3 are a well-scoped project. The payoff is the first dynamical check of the $\mathrm{DLCQ}^n/\mathrm{DLCQ}^{n+1}$ conjecture beyond $n = 0$. It would also turn "non-Lorentzian holography" from a list of limits into a correspondence with matched observables, and it addresses the paper's main weakness directly.

By comparison, quantum or non-abelian versions of the p-brane $T\bar{T}$ flows would have high impact, but they are much less certain to succeed: there is no factorisation theorem for $p > 1$, and the non-abelian DBI action is not known to all orders.